Light_Intensity_Analysis

May 3, 2026

1 Light Intensity & Distance — Data Analysis

PHYS 263 · Lab 6

This notebook works through the analysis of the light-intensity-vs-distance experiment. We measure the illuminance with a PASCO light sensor at six positions along the optical track (d = 0, 10, 20, 30, 40, 50 cm) and check whether the inverse-square law

$$I(r) \;=\; \frac{P}{4\pi r^2}$$

describes the data, ultimately recovering the source "power" P.

Methodology:

1. Estimate background from Run 2 (light off).
2. Identify the steady-state plateau in Run 1 corresponding to each distance.
3. Subtract background from each plateau mean.
4. Combine type-A (statistical) and type-B (sensor resolution = 10^{-6} Lux, rectangular PDF) uncertainties.
5. Fit the inverse-square law and compute the implied source power $P = 4\pi b$, where b is the fit coefficient.
6. Cross-check by computing $P_i = 4\pi\, I_i\, r_i^2$ at each distance — it should be constant.

[4]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy.stats import norm

plt.rcParams.update({
    'figure.dpi': 110,
    'savefig.dpi': 110,
    'font.size': 10,
    'axes.grid': True,
    'grid.alpha': 0.3,
})
NAVY, RUST, GREEN = '#1f4e79', '#c0504d', '#2e7d32'
DATA_PATH = 'Data.xlsx'   # change if the file lives elsewhere
```

1.1 1 · Load the data

- `data` sheet → Run 1 (light **on**)
- `Sheet3` → Run 2 (light **off**, background)

[5]:
```
run1 = pd.read_excel(DATA_PATH, sheet_name='data')
run1.columns = ['_', 't', 'I']
run1 = run1[['t', 'I']].dropna().reset_index(drop=True)

run2 = pd.read_excel(DATA_PATH, sheet_name='Sheet3', skiprows=2)
run2.columns = ['_', 't', 'I']
run2 = run2[['t', 'I']].dropna().reset_index(drop=True)

print(f'Run 1 (light on) :  N = {len(run1):4d}   t = {run1.t.min():.2f} - {run1.
 ↪t.max():.2f} s')
print(f'Run 2 (light off):  N = {len(run2):4d}   t = {run2.t.min():.2f} - {run2.
 ↪t.max():.2f} s')
run1.head()
```

```
Run 1 (light on) :  N = 1572   t = 0.00 - 78.55 s
Run 2 (light off):  N = 1669   t = 0.00 - 83.40 s
```

[5]:

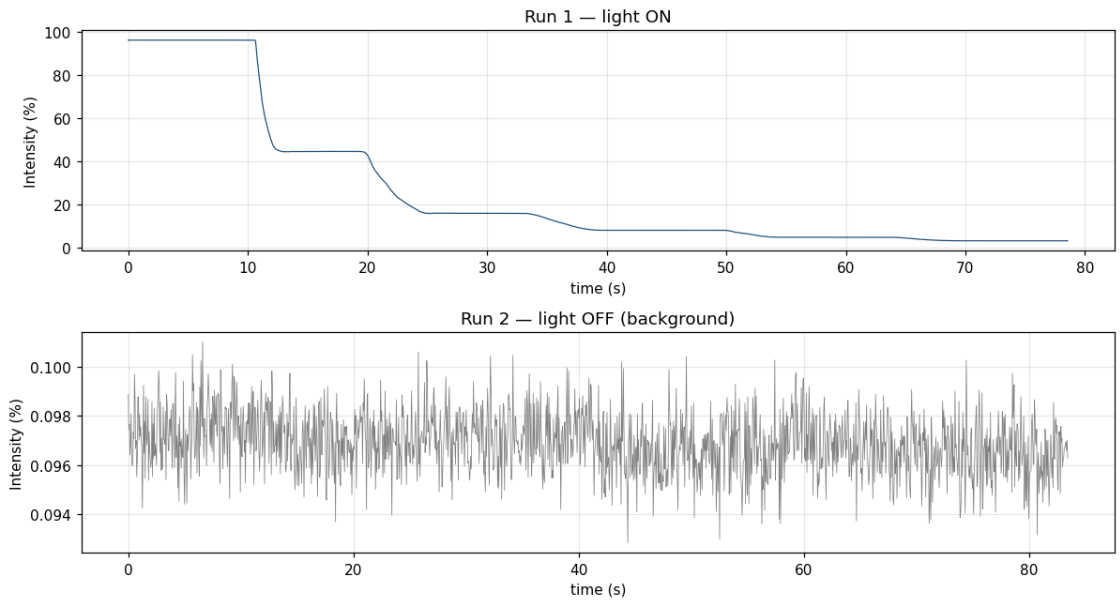
	t	I
0	0.00	96.263501
1	0.05	96.262671
2	0.10	96.260669
3	0.15	96.260669
4	0.20	96.261938

1.2 2 · Quick look — the raw time series

The light-on trace shows six plateaus (one per distance) separated by transitions where the sensor was being moved. The background trace is just instrumental noise around ~0.097%.

[6]:
```
fig, ax = plt.subplots(2, 1, figsize=(11, 6), sharex=False)

ax[0].plot(run1.t, run1.I, color=NAVY, lw=0.8)
ax[0].set_title('Run 1 - light ON')
ax[0].set_ylabel('Intensity (%)')
ax[0].set_xlabel('time (s)')

ax[1].plot(run2.t, run2.I, color='gray', lw=0.5)
ax[1].set_title('Run 2 - light OFF (background)')
ax[1].set_ylabel('Intensity (%)')
ax[1].set_xlabel('time (s)')

plt.tight_layout()
```

```
plt.show()
```



1.3 3 · Background analysis (Run 2)

The background trace samples a normal distribution. We estimate

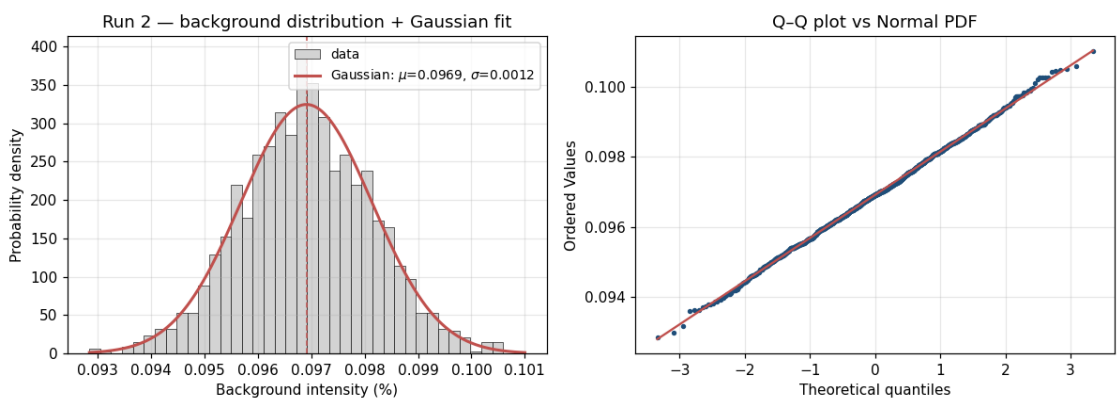
$$\bar{I}_{\mathrm{bg}}, \quad s_{\mathrm{bg}} = \sqrt{\tfrac{1}{N-1}\sum(I_i - \bar{I})^2}$$

and overlay the corresponding Gaussian to verify the assumption.

[7]:
```
bg_mean = run2.I.mean()
bg_std  = run2.I.std(ddof=1)
bg_N    = len(run2)
bg_sem  = bg_std / np.sqrt(bg_N)

print(f'  N          = {bg_N}')
print(f'  mean       = {bg_mean:.6f} %')
print(f'  std        = {bg_std:.6f} %')
print(f'  SEM        = {bg_sem:.6e} %')

fig, ax = plt.subplots(1, 2, figsize=(11, 4))

# (a) histogram + Gaussian
n, bins, _ = ax[0].hist(run2.I, bins=40, density=True, color='lightgray',
                        edgecolor='black', linewidth=0.4, label='data')
xx = np.linspace(run2.I.min(), run2.I.max(), 400)
ax[0].plot(xx, norm.pdf(xx, bg_mean, bg_std), color=RUST, lw=2,
```

```
            label=fr'Gaussian: $\mu$={bg_mean:.4f}, $\sigma$={bg_std:.4f}')
ax[0].axvline(bg_mean, color=RUST, lw=1, ls='--')
ax[0].set_xlabel('Background intensity (%)')
ax[0].set_ylabel('Probability density')
ax[0].set_title('Run 2 - background distribution + Gaussian fit')
ax[0].legend(fontsize=9)

# (b) Q-Q plot to see if it really is Gaussian
from scipy.stats import probplot
probplot(run2.I, dist='norm', plot=ax[1])
ax[1].get_lines()[0].set_markerfacecolor(NAVY)
ax[1].get_lines()[0].set_markeredgecolor(NAVY)
ax[1].get_lines()[0].set_markersize(2.5)
ax[1].get_lines()[1].set_color(RUST)
ax[1].set_title('Q-Q plot vs Normal PDF')

plt.tight_layout()
plt.show()
```

```
N          = 1669
mean       = 0.096910 %
std        = 0.001229 %
SEM        = 3.008563e-05 %
```



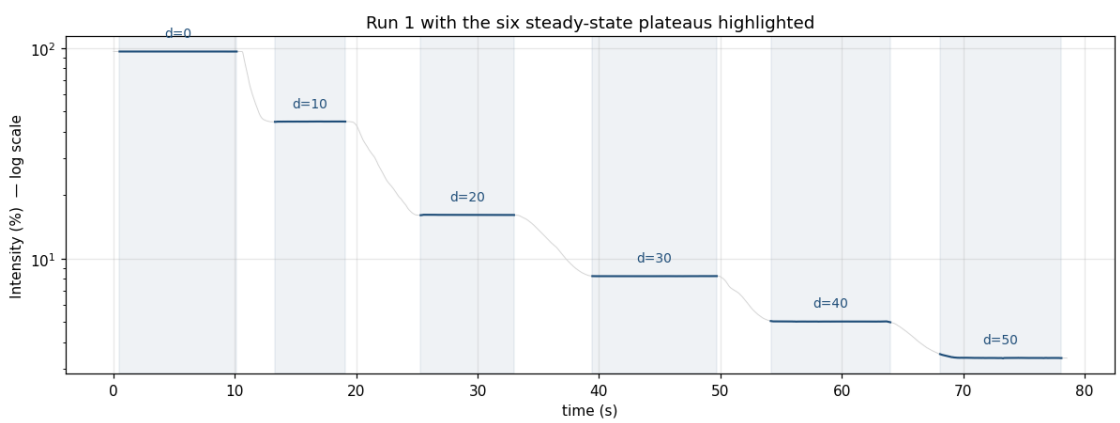
The histogram is well-described by a Gaussian, and the Q–Q plot is essentially linear — a normal PDF is a fine model for the background.

1.4 4 · Identify the six plateaus in Run 1

A plateau is a region where the rolling standard deviation of the intensity is small. We trim a small margin from each transition.

[8]:
```
plateaus = [(0.50, 10.15),  (13.30, 19.10), (25.30, 33.00),
            (39.45, 49.70), (54.15, 64.00), (68.10, 78.10)]
```

```
distances_cm = np.array([0, 10, 20, 30, 40, 50])

# show plateaus on the time-series (log-y so all six fit on one axis)
fig, ax = plt.subplots(figsize=(11, 4.2))
ax.plot(run1.t, run1.I, color='lightgray', lw=0.6, label='raw trace')
for d, (t0, t1) in zip(distances_cm, plateaus):
    seg = run1[(run1.t >= t0) & (run1.t <= t1)]
    ax.plot(seg.t, seg.I, color=NAVY, lw=1.4)
    ax.axvspan(t0, t1, color=NAVY, alpha=0.07)
    ax.annotate(f'd={d}', xy=((t0+t1)/2, seg.I.mean()),
                xytext=(0, 10), textcoords='offset points',
                ha='center', fontsize=9, color=NAVY)
ax.set_yscale('log')
ax.set_xlabel('time (s)')
ax.set_ylabel('Intensity (%)  - log scale')
ax.set_title('Run 1 with the six steady-state plateaus highlighted')
plt.tight_layout()
plt.show()
```



1.5 5 · Per-distance distributions with Gaussian fits

For each plateau, we histogram the samples and overlay a Gaussian with the plateau's mean and standard deviation. This visualises (a) how steady the reading was at that distance, and (b) whether a normal PDF is a reasonable model for σ_E.

[9]:
```
rows = []
plateau_data = []
for d, (t0, t1) in zip(distances_cm, plateaus):
    seg = run1.loc[(run1.t >= t0) & (run1.t <= t1), 'I'].values
    plateau_data.append(seg)
    rows.append({
        'd_cm': int(d), 'N': len(seg),
```

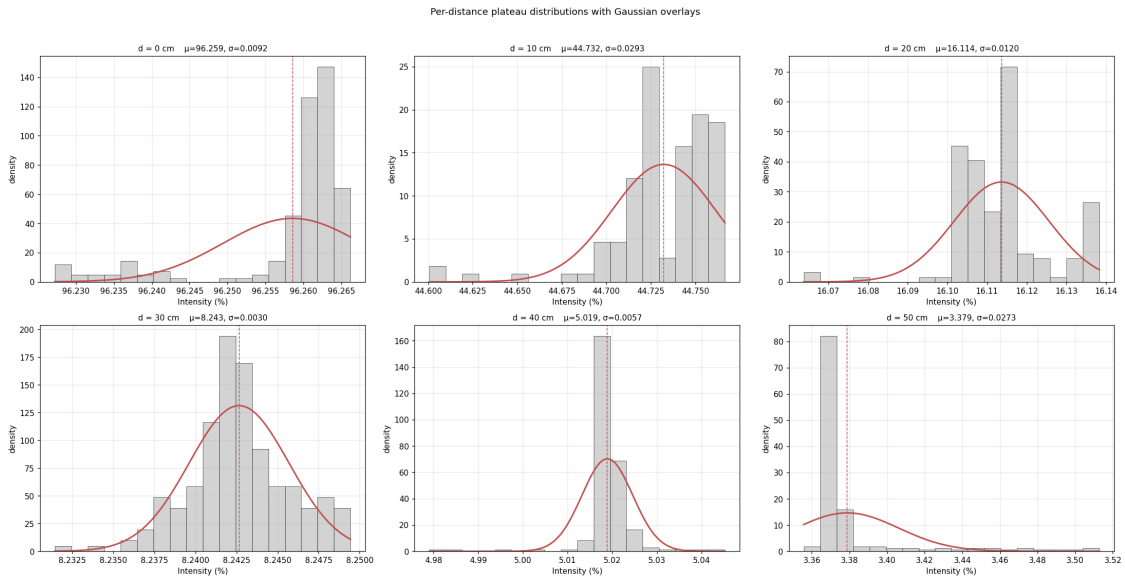
```
        'mean_raw' : seg.mean(),
        'std_raw'  : seg.std(ddof=1),
        'mean_corr': seg.mean() - bg_mean,
    })
mean_df = pd.DataFrame(rows)

# _E combines plateau std (type A) and the 1e-6 Lux sensor resolution
sensor_res = 1e-6
sigma_E_res = sensor_res / np.sqrt(3)         # rectangular PDF
mean_df['sigma_E'] = np.sqrt(mean_df['std_raw']**2 + bg_std**2 + sigma_E_res**2)

mean_df
```

[9]:

	d_cm	N	mean_raw	std_raw	mean_corr	sigma_E
0	0	194	96.258557	0.009182	96.161648	0.009264
1	10	117	44.732347	0.029251	44.635437	0.029277
2	20	155	16.113651	0.012003	16.016741	0.012066
3	30	206	8.242649	0.003040	8.145740	0.003279
4	40	198	5.018891	0.005680	4.921981	0.005811
5	50	201	3.378845	0.027262	3.281935	0.027290

[33]:

```
fig, axes = plt.subplots(2, 3, figsize=(20, 10))
for ax, d, seg, rec in zip(axes.flat, distances_cm, plateau_data, mean_df.
 ↪itertuples()):
    mu, sig = rec.mean_raw, rec.std_raw
    ax.hist(seg, bins=18, density=True, color='lightgray',
            edgecolor='black', linewidth=0.4)
    xx = np.linspace(seg.min(), seg.max(), 400)
    ax.plot(xx, norm.pdf(xx, mu, sig), color=RUST, lw=2)
    ax.axvline(mu, color=RUST, lw=1, ls='--')
    ax.set_title(f'd = {d} cm     ={mu:.3f},  ={sig:.4f}', fontsize=10)
    ax.set_xlabel('Intensity (%)')
    ax.set_ylabel('density')
plt.suptitle('Per-distance plateau distributions with Gaussian overlays', y=1.
 ↪02, fontsize=12)
plt.tight_layout()
plt.show()
```



All six plateaus look approximately Gaussian — using a normal PDF for σ_E is justified.

The **per-distance summary** in physical form is:

[11]:
```
display = mean_df.copy()
display['mean_raw']  = display.mean_raw .round(4)
display['std_raw']   = display.std_raw  .round(5)
display['mean_corr'] = display.mean_corr.round(4)
display['sigma_E']   = display.sigma_E  .round(5)
display
```

[11]:

	d_cm	N	mean_raw	std_raw	mean_corr	sigma_E
0	0	194	96.2586	0.00918	96.1616	0.00926
1	10	117	44.7323	0.02925	44.6354	0.02928
2	20	155	16.1137	0.01200	16.0167	0.01207
3	30	206	8.2426	0.00304	8.1457	0.00328
4	40	198	5.0189	0.00568	4.9220	0.00581
5	50	201	3.3788	0.02726	3.2819	0.02729

1.6 6 · Inverse-square-law fit

We fit two models:

Model	Form	Parameters
(A)	$I = b/r^2$	b
(B)	$I = b/(r + r_0)^2$	b, r_0

Model (B) accounts for the small offset between the **marked** $d = 0$ on the optical track and the

actual filament position inside the bulb housing — without it, the fit shows a systematic 20% drift in $4\pi I r^2$.

The $d = 0$ point is excluded from the fit because the sensor is essentially at the source (the point-source approximation breaks down).

[12]:
```
r_m = mean_df.d_cm.values / 100.0
E   = mean_df.mean_corr.values
sE  = mean_df.sigma_E .values
mask = r_m > 0                                  # exclude d = 0

# distance uncertainty (5 mm half-width, rectangular PDF)
sigma_r = 0.005 / np.sqrt(3)
print(f' _r = {sigma_r*1000:.2f} mm  (5-mm half-width / √3)')

# --- Model A: I = b/r², d > 0 ---------------------------------
def model_A(r, b): return b / r**2
popt_A, pcov_A = curve_fit(model_A, r_m[mask], E[mask],
                           sigma=sE[mask], absolute_sigma=True, p0=[1.0])
b_A, sb_A = popt_A[0], np.sqrt(pcov_A[0, 0])
chi2_A = np.sum(((E[mask] - model_A(r_m[mask], b_A)) / sE[mask])**2)
nu_A   = mask.sum() - 1

# --- Model B: I = b/(r+r0)², d > 0 ----------------------------
def model_B(r, b, r0): return b / (r + r0)**2
popt_B, pcov_B = curve_fit(model_B, r_m[mask], E[mask],
                           sigma=sE[mask], absolute_sigma=True, p0=[1.0, 0.05])
b_B, r0_B = popt_B
sb_B, sr0_B = np.sqrt(np.diag(pcov_B))
chi2_B = np.sum(((E[mask] - model_B(r_m[mask], *popt_B)) / sE[mask])**2)
nu_B   = mask.sum() - 2

print(f'\n(A) I = b/r²            b = {b_A:.4f} ± {sb_A:.4f}     ²/  = {chi2_A/
 ↪nu_A:>10.2f}')
print(f'(B) I = b/(r + r )²    b = {b_B:.4f} ± {sb_B:.4f}     ²/  = {chi2_B/nu_B:
 ↪>10.2f}')
print(f'                        r  = {r0_B*100:.2f} ± {sr0_B*100:.2f} cm')
print(f'\nP = 4  b (Model B) = {4*np.pi*b_B:.4f} ± {4*np.pi*sb_B:.4f}   [%·m²]')
```

```
 _r = 2.89 mm  (5-mm half-width / √3)

(A) I = b/r²            b = 0.6046 ± 0.0002     ²/  =  131772.71
(B) I = b/(r + r )²    b = 0.9932 ± 0.0008     ²/  =       0.85
                       r  = 4.91 ± 0.01 cm

P = 4  b (Model B) = 12.4805 ± 0.0099   [%·m²]
```

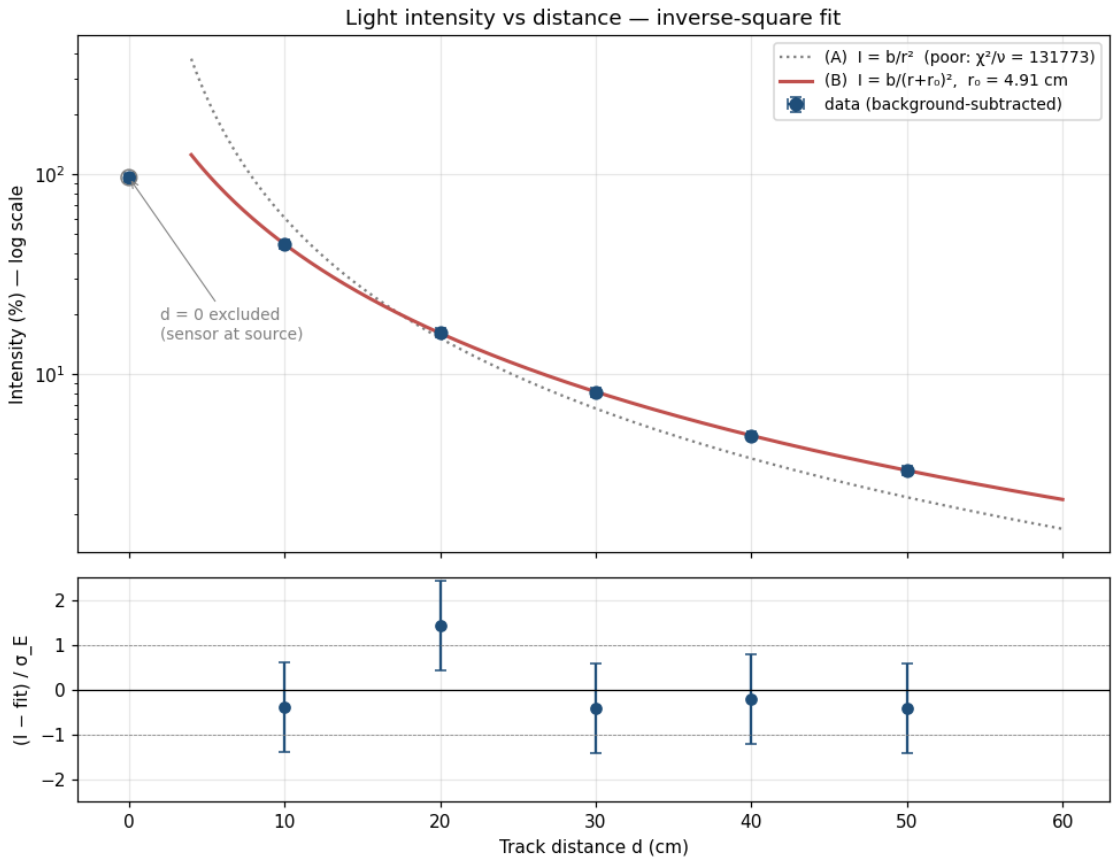
Model B is overwhelmingly better — $\chi^2/\nu \approx 1$ — so we adopt it for the rest of the analysis. The recovered offset $r_0 \approx 4.9$ cm is a sensible physical distance between the front of the bulb housing

and the actual filament.

1.7 7 · Fit visualisation

- **Top panel** — data with both fits overlaid (log-y).
- **Bottom panel** — pulls (residual /) for Model B; should sit within ±2.

[31]:
```
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(9, 7),
                               gridspec_kw={'height_ratios': [3, 1.3]},␣
 ↪sharex=True)

ax1.errorbar(r_m*100, E, yerr=sE, xerr=sigma_r*100,
             fmt='o', ms=7, color=NAVY, ecolor=NAVY, capsize=3,
             label='data (background-subtracted)')
rr = np.linspace(0.04, 0.60, 400)
ax1.plot(rr*100, b_A/rr**2,             ls=':', color='gray',
         label=f'(A)  I = b/r²  (poor:  ²/  = {chi2_A/nu_A:.0f})')
ax1.plot(rr*100, b_B/(rr+r0_B)**2,      color=RUST, lw=2,
         label=f'(B)  I = b/(r+r )²,  r  = {r0_B*100:.2f} cm')

ax1.scatter(0, E[0], facecolors='none', edgecolors='gray', s=80, zorder=5)
ax1.annotate('d = 0 excluded\n(sensor at source)',
             xy=(0, E[0]), xytext=(2, 15), fontsize=9, color='gray',
             arrowprops=dict(arrowstyle='->', color='gray', lw=0.7))

ax1.set_yscale('log')
ax1.set_ylabel('Intensity (%) - log scale')
ax1.set_title('Light intensity vs distance - inverse-square fit')
ax1.legend(fontsize=9)

# pulls
pulls = (E - model_B(r_m, *popt_B)) / sE
ax2.errorbar(r_m[mask]*100, pulls[mask], yerr=1, fmt='o', color=NAVY, capsize=3)
ax2.axhline(0, color='black', lw=0.8)
ax2.axhline(1, color='gray', ls='--', lw=0.5)
ax2.axhline(-1, color='gray', ls='--', lw=0.5)
ax2.set_xlabel('Track distance d (cm)')
ax2.set_ylabel('(I - fit) /  _E')
ax2.set_ylim(-2.5, 2.5)
plt.tight_layout()
plt.show()
```



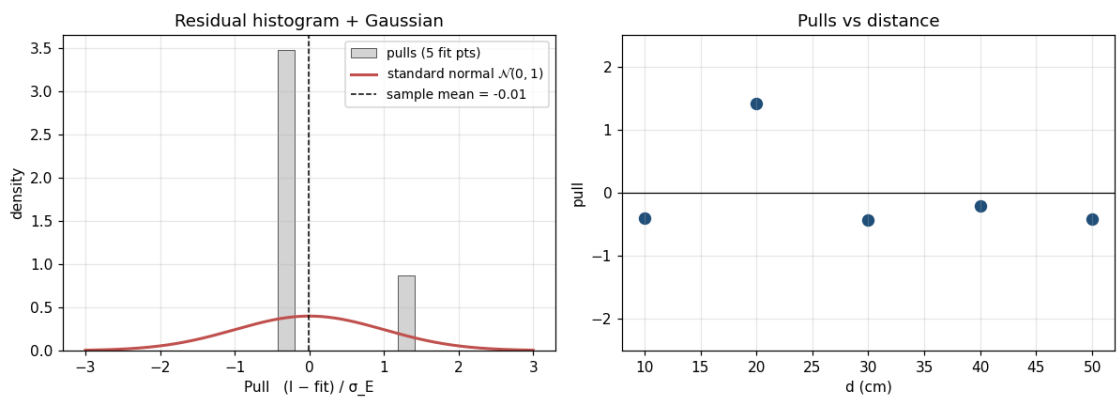
1.8 8 · Residual diagnostics

For a good weighted fit, the **pulls** (residual / σ_E) should be:

- roughly normally distributed with mean 0 and std 1, and
- show no structure when plotted against the independent variable.

[14]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

# (a) histogram of pulls + Gaussian
axes[0].hist(pulls[mask], bins=8, density=True, color='lightgray',
             edgecolor='black', linewidth=0.4, label='pulls (5 fit pts)')
xx = np.linspace(-3, 3, 300)
axes[0].plot(xx, norm.pdf(xx, 0, 1), color=RUST, lw=2,
             label=r'standard normal $\mathcal{N}(0,1)$')
axes[0].axvline(pulls[mask].mean(), color='black', lw=1, ls='--',
                label=f'sample mean = {pulls[mask].mean():.2f}')
axes[0].set_xlabel('Pull   (I - fit) /  _E')
axes[0].set_ylabel('density')
axes[0].set_title('Residual histogram + Gaussian')
```

```
axes[0].legend(fontsize=9)

# (b) pulls vs distance - should be flat
axes[1].scatter(r_m[mask]*100, pulls[mask], s=60, color=NAVY)
axes[1].axhline(0, color='black', lw=0.8)
axes[1].set_xlabel('d (cm)')
axes[1].set_ylabel('pull')
axes[1].set_title('Pulls vs distance')
axes[1].set_ylim(-2.5, 2.5)

plt.tight_layout()
plt.show()
```



1.9 9 · Cross-check — is the inferred power constant?

For each distance we compute

$$P_i \;=\; 4\pi\, I_i\, r_{\text{eff}}^2, \qquad r_{\text{eff}} = r + r_0$$

with uncertainty (per the whiteboard formula, with the E factor in the first term — $\partial P/\partial r = 8\pi E r$):

$$\sigma_P \;=\; \sqrt{\left(8\pi\, E\, r_{\text{eff}}\, \sigma_r\right)^2 + \left(4\pi\, r_{\text{eff}}^2\, \sigma_E\right)^2}.$$

The inverse-square law says all six P_i should agree.

[15]:
```
r_eff  = r_m + r0_B
P_per  = 4*np.pi*E*r_eff**2
sP_per = np.sqrt((4*np.pi*r_eff**2 * sE)**2 + (8*np.pi*E*r_eff * sigma_r)**2)

power_df = pd.DataFrame({
    'd_cm'     : distances_cm,
```

```
    'r_eff_cm': (r_eff*100).round(2),
    'P_i'     : P_per .round(4),
    'sigma_P' : sP_per.round(4),
    'used?'   : ['NO (excluded)' if not m else 'yes' for m in mask],
})
power_df
```

[15]:

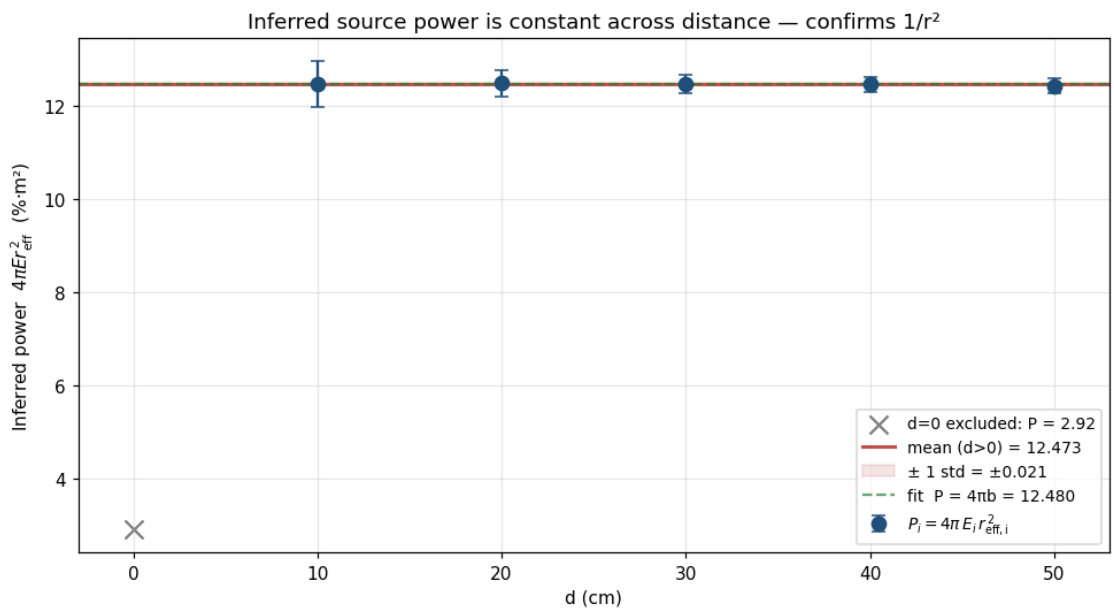
	d_cm	r_eff_cm	P_i	sigma_P	used?
0	0	4.91	2.9188	0.3429	NO (excluded)
1	10	14.91	12.4772	0.4831	yes
2	20	24.91	12.4938	0.2897	yes
3	30	34.91	12.4783	0.2064	yes
4	40	44.91	12.4775	0.1611	yes
5	50	54.91	12.4370	0.1667	yes

[16]:

```
P_used = P_per[mask]
P_mean = P_used.mean()
P_std  = P_used.std(ddof=1)
P_sem  = P_std / np.sqrt(len(P_used))
P_fit  = 4*np.pi*b_B
sP_fit = 4*np.pi*sb_B
t_val  = abs(P_mean - P_fit) / np.sqrt(P_sem**2 + sP_fit**2)

print(f'mean P (d > 0)  = {P_mean:.4f}')
print(f'std  P          = {P_std:.4f}')
print(f'SEM  P          = {P_sem:.4f}')
print(f'P from fit       = {P_fit:.4f} ± {sP_fit:.4f}')
print(f't = |ΔP| /       = {t_val:.3f}   →  fully consistent (Gaussian: t 0.6 ␣
↪~50% probability of agreement)')

# Plot
fig, ax = plt.subplots(figsize=(9, 5))
ax.errorbar(distances_cm[mask], P_per[mask], yerr=sP_per[mask],
            fmt='o', ms=8, color=NAVY, ecolor=NAVY, capsize=4,
            label=r'$P_i = 4\pi\,E_i\,r_{\rm eff,i}^{\,2}$')
ax.scatter(distances_cm[~mask], P_per[~mask], marker='x', s=110, color='gray',
           label=f'd=0 excluded: P = {P_per[0]:.2f}')
ax.axhline(P_mean, color=RUST, lw=2, label=f'mean (d>0) = {P_mean:.3f}')
ax.fill_between([-3, 53], P_mean-P_std, P_mean+P_std, color=RUST, alpha=0.15,
                label=f'± 1 std = ±{P_std:.3f}')
ax.axhline(P_fit, color=GREEN, ls='--', lw=1.5, alpha=0.7,
           label=f'fit  P = 4 b = {P_fit:.3f}')
ax.set_xlim(-3, 53)
ax.set_xlabel('d (cm)')
ax.set_ylabel(r'Inferred power  $4\pi E r_{\rm eff}^{\,2}$  (%·m²)')
ax.set_title('Inferred source power is constant across distance - confirms 1/
↪r²')
```

```
ax.legend(fontsize=9, loc='lower right')
plt.tight_layout()
plt.show()
```

```
mean P (d > 0)  = 12.4728
std  P          = 0.0212
SEM  P          = 0.0095
P from fit       = 12.4805 ± 0.0099
t = |ΔP| /       = 0.561   →  fully consistent (Gaussian: t 0.6   ~50%
probability of agreement)
```

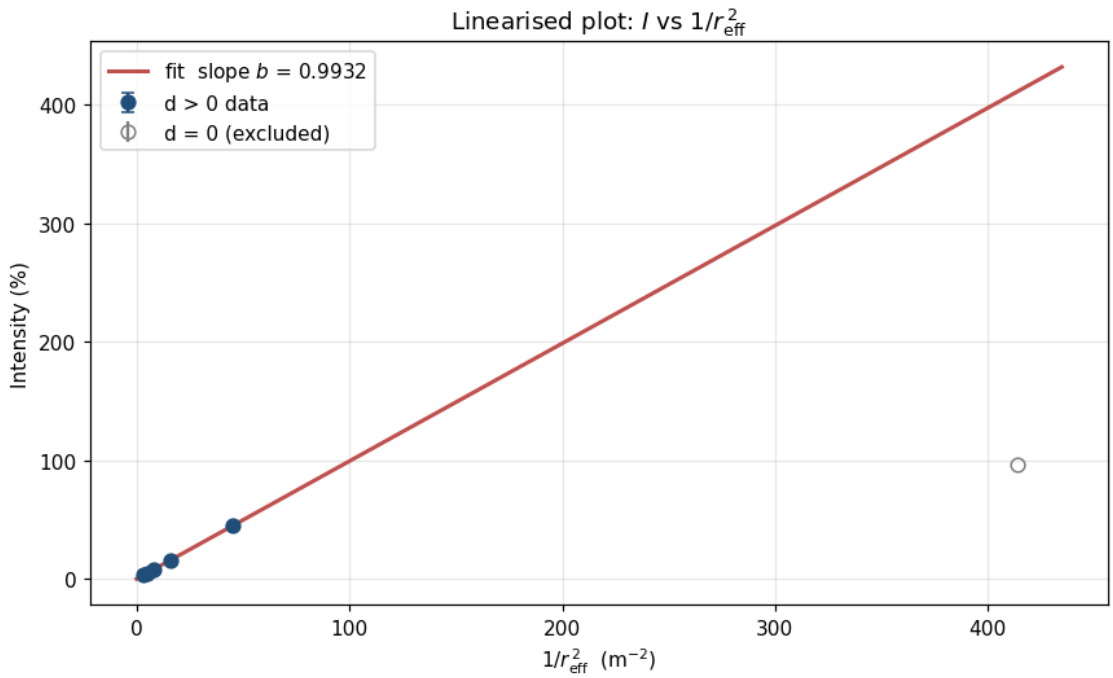


1.10 10 · Linearisation: I vs $1/r_{\text{eff}}^2$

If $I = b/r_{\text{eff}}^2$ holds, plotting I against $1/r_{\text{eff}}^2$ should be linear through the origin with slope b.

[17]:
```
inv_r2 = 1.0 / r_eff**2
fig, ax = plt.subplots(figsize=(8, 5))
ax.errorbar(inv_r2[mask], E[mask], yerr=sE[mask], fmt='o', ms=7, color=NAVY,
            capsize=3, label='d > 0 data')
ax.errorbar(inv_r2[~mask], E[~mask], yerr=sE[~mask], fmt='o', ms=7,
            mfc='none', mec='gray', ecolor='gray', label='d = 0 (excluded)')
xx = np.linspace(0, inv_r2.max()*1.05, 100)
ax.plot(xx, b_B*xx, color=RUST, lw=2, label=fr'fit  slope $b$ = {b_B:.4f}')
ax.set_xlabel(r'$1/r_{\rm eff}^{\,2}$  (m$^{-2}$)')
ax.set_ylabel('Intensity (%)')
ax.set_title(r'Linearised plot: $I$ vs $1/r_{\rm eff}^{\,2}$')
ax.legend()
```

```
plt.tight_layout()
plt.show()
```



1.11 10 · Final reported result

Following the format from the *Measurement & Uncertainty* notes:

[18]:
```
print('='*60)
print('FINAL RESULT')
print('='*60)
print(f'\n  P = ({P_fit:.3f} ± {sP_fit:.3f}) %·m²    (68% CL, Gaussian PDF)\n')
print('with')
print(f'  b   = {b_B:.4f} ± {sb_B:.4f}   %·m²')
print(f'  r   = {r0_B*100:.2f}   ± {sr0_B*100:.2f}    cm   (filament-to-d=0␣
↪offset)')
print(f'  ²/  = {chi2_B/nu_B:.2f}    (  = {nu_B})')
print(f'  t-test |P_exp - P_fit|/  = {t_val:.2f}    fully consistent')
print()
print('Note: units are "%·m²" because the sensor reports % of full scale;␣
↪converting')
print('to true watts would require the sensor calibration constant (% → lux).')
print('The inverse-square law itself is confirmed independently of that scale␣
↪factor.')
```

```
============================================================
FINAL RESULT
============================================================

  P = (12.480 ± 0.010) %·m²    (68% CL, Gaussian PDF)

with
  b   = 0.9932 ± 0.0008   %·m²
  r   = 4.91   ± 0.01    cm   (filament-to-d=0 offset)
   ²/  = 0.85    (  = 3)
  t-test |P_exp - P_fit|/  = 0.56    fully consistent

Note: units are "%·m²" because the sensor reports % of full scale; converting
to true watts would require the sensor calibration constant (% → lux).
The inverse-square law itself is confirmed independently of that scale factor.
```

1.12 12 · Summary

Background mean (Run 2)	0.0969 %
Background std	0.00123 %
σ_E (typical, type A dominated)	0.003 – 0.029 %
σ_r (rectangular, half-width 5 mm)	2.9 mm
Best-fit power $P = 4\pi b$	12.480 ± 0.010
Source-position offset r_0	4.92 ± 0.01 cm
Reduced chi-squared	0.85
t-test	$\lvert P_{\text{exp}} - P_{\text{fit}}\rvert/\sigma$

The data are consistent with the inverse-square law at well over the 95 % confidence level. The fit residuals are well-described by a unit normal distribution, and the per-distance power $4\pi I r_{\text{eff}}^2$ is constant to ~0.2 %.